Analysis of NBS: Engineering Performance

CUNY EES 79903

Seminar on Nature-Based Solutions

Kyle McKay

October 20, 2026

Lecture Overview:

- 415-420 – Housekeeping
- 420-445 – Lecture: NBS engineering guidance development
- 445-615 – Discussion: Corwin et al. (2026), Bredes et al. (2024), Esquivel-Muelbert et al. (2026)

	Leader #1	Leader #2
~~Sep 8: NBS fundamentals~~	~~Grace~~	~~Jesse~~
~~Sep 15: NBS fundamentals~~	~~Michelle~~	~~Shahela~~
~~Sep 22: Structured decision making and community engagement~~	~~Jesse~~	~~Aurelia~~
~~Sep 29: Analysis – ecological~~	~~Carol~~	~~Julia~~
~~Oct 6: Analysis – co-benefits~~	~~Shahela~~	~~Aurelia~~
Oct 20: Analysis – engr performance	Grace	Michelle
Oct 27: Analysis – engr performance	Abdellah	Julia
Nov 3: Alternatives analysis	Aurelia	Abdellah
Nov 10: Performance-based design	Aaryan	Michelle
Nov 17: Constructability	Grace	Aaryan
Nov 24: Performance monitoring	Shahela	Jesse
Dec 1: Performance monitoring	Julia	Carol
Dec 8: Life cycle analysis	Aaryan	Abdellah
Dec 15: Course synthesis	Kyle	Brett

Project Reminder

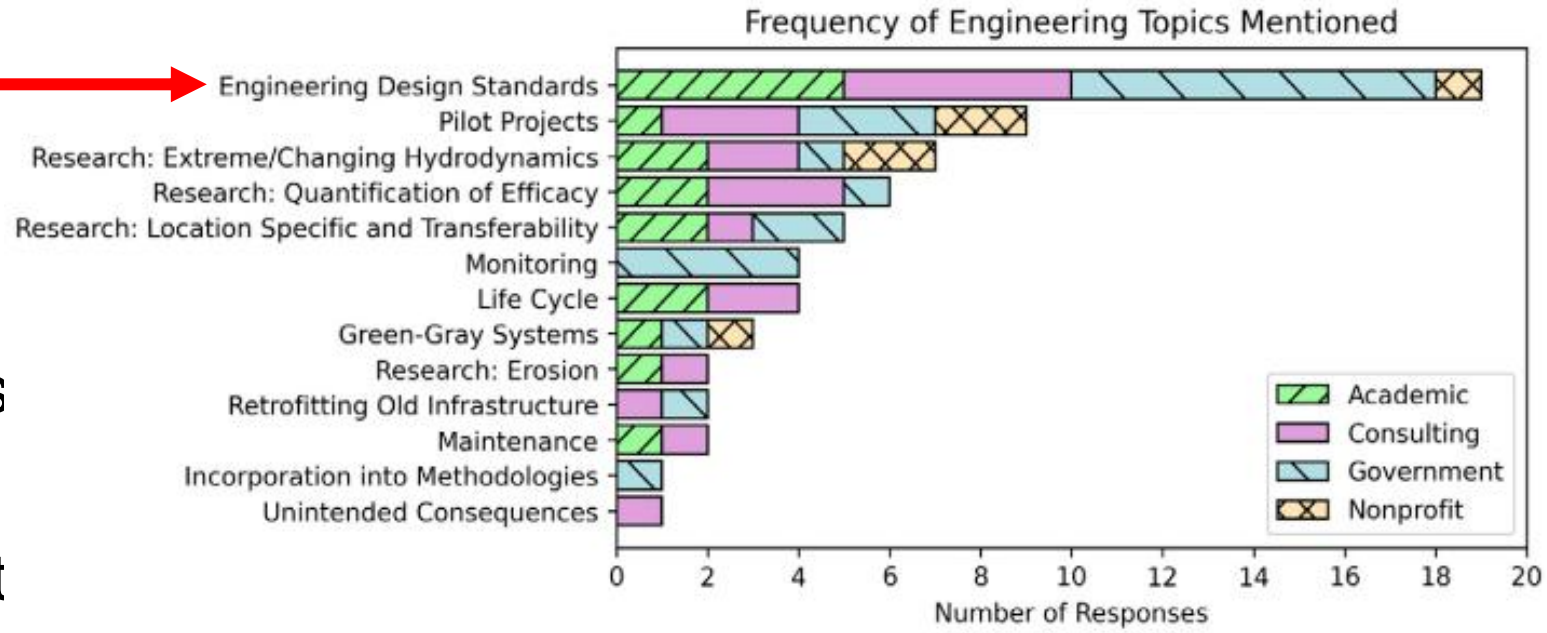
- Oct 20 – Recommended timeframe to compile initial references
- Nov 3 – Recommended timeframe for outlining ideas
- Nov 17 – Initial submission of traceable account
- Nov 24 – Review of traceable account by classmates
- Dec 8 – Final submission of traceable account
- Dec 15 – Course synthesis (McKay)

Topic	First option
Qualitative foundations of project planning	Jesse
Facilitation, engagement, and teaming	Aurelia
Ecological analysis of NBS	Julia
Economic analysis of NBS	
Co-benefits analysis of NBS	
Engineering analysis of NBS	Abdellah
Alternatives analysis and structured decision making	
Performance-based design	Aaryan
Constructability	Michelle
Performance monitoring	
Project establishment	
Asset management	Grace

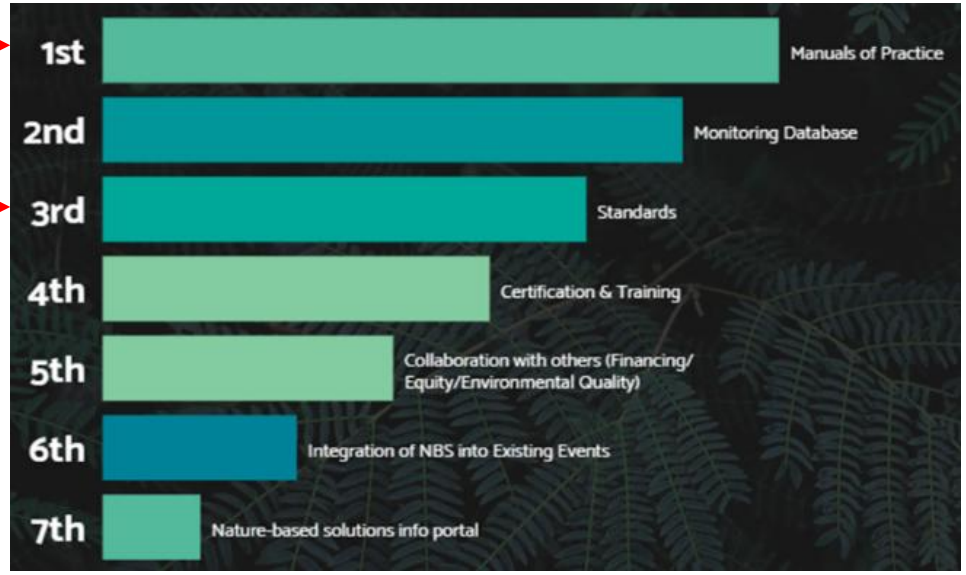
Developing a nature-based solutions manual of practice with the American Society of Civil Engineers

Why do engineers need guidance?

- Guidance establishes “standard of care”
- Rapidly evolving pract and non-traditional materials
- NBS design requires different trade-offs than traditionally encountered
- Engineers say that’s (part) of what limits use of NBS



Ostrow et al. (2022). Frequency of categories of future engineering needs for emergent vegetation implementation cited by experts in survey responses (N=32).



ASCE Inspire (2023). Prioritization of potential ASCE initiatives to develop NBS engineering resources.

What is engineering guidance?

A suite of resources that:

- Span a spectrum of formal to informal content
- Are developed by many institutions
- Nest together to collectively provide engineers with defensible processes
- Dynamically capture the state-of-the-practice

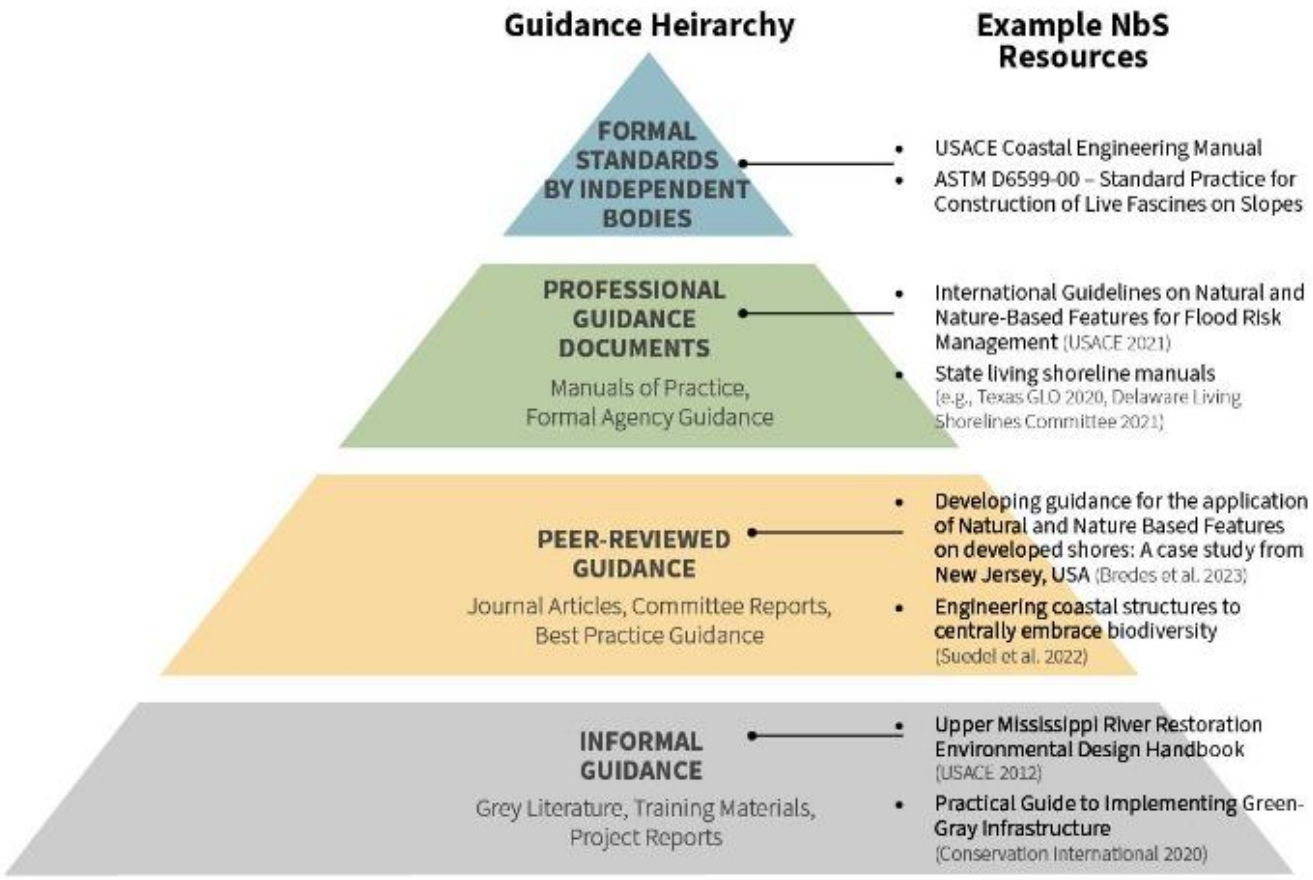
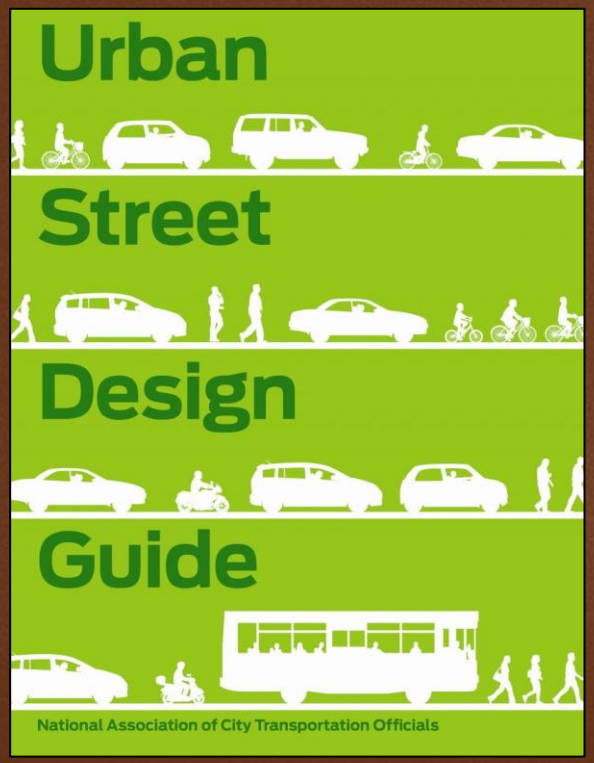


Figure: Corwin et al. (2026, ASCE Open)

How does guidance fit together?

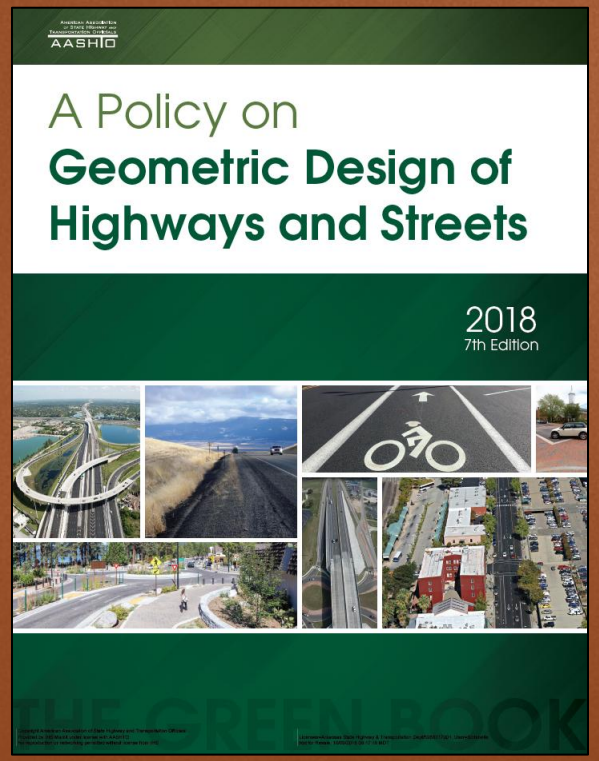
Interdisciplinary context:

- Funding strategy
- Systems planning
- Stakeholder engagement
- Multi-purpose outcomes



Engineering foundations:

- Key concepts
- Engineer's role
- Level of service
- Design process



Engineering practice:

- Analytical processes
- Problem-centric design
- Secondary features
- State and regional guides

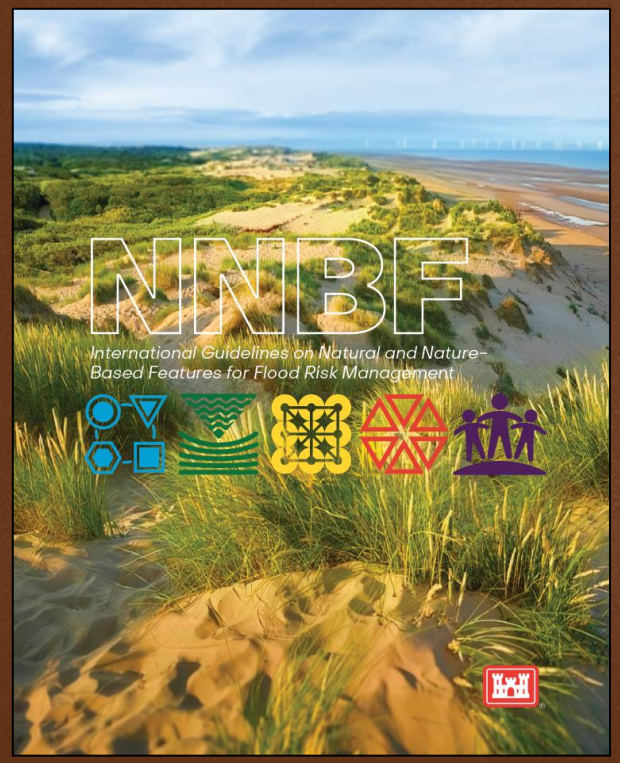


WOOLPERT

How does guidance fit together?

Interdisciplinary context:

- Funding strategy
- Systems planning
- Stakeholder engagement
- Multi-purpose outcomes



Engineering foundations:

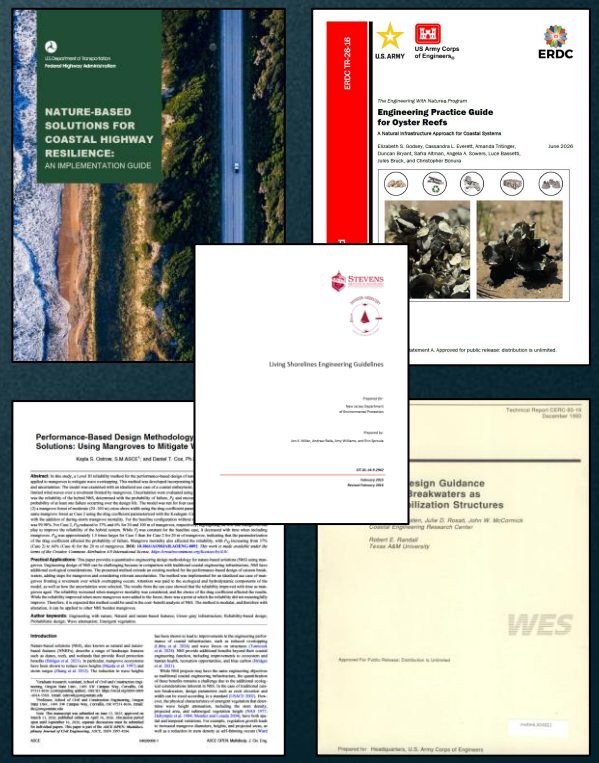
- Key concepts
- Engineer's role
- Level of service
- Design process

ASCE
AMERICAN SOCIETY OF CIVIL ENGINEERS

Natural Infrastructure Manual of Practice

Engineering practice:

- Analytical processes
- Problem-centric design
- Secondary features
- State and regional guides



WOOLPERT

What are the needs for an NBS manual?

Translation of NBS concepts into engineering lexicon





Design criteria
Level of service
Project teaming
Assets
O&M
Standards
Code

Connecting NBS to multiple engineering sectors

Coastal

Geotechnical

Structures

ASCE
AMERICAN SOCIETY OF CIVIL ENGINEERS

Natural Infrastructure Manual of Practice

Transportation

Water Resources

Environmental Engineering

Reducing redundancy and freeing other NBS guidance to focus on details

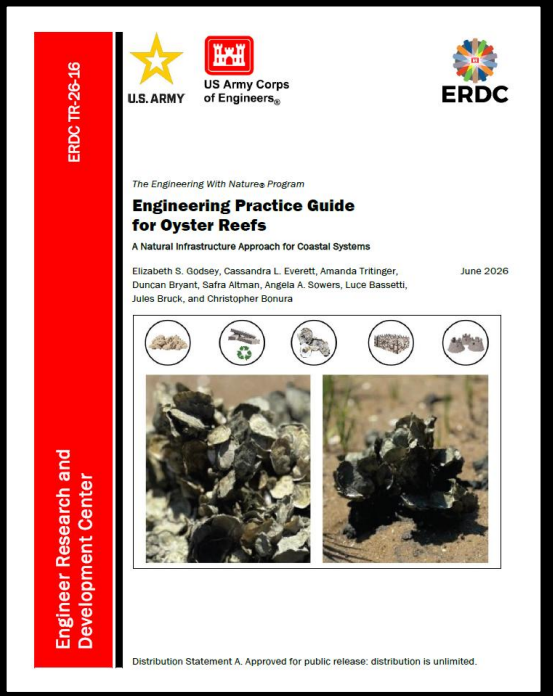


Table 17. Fitting coefficients for Provan et al. (2024) stability equations.

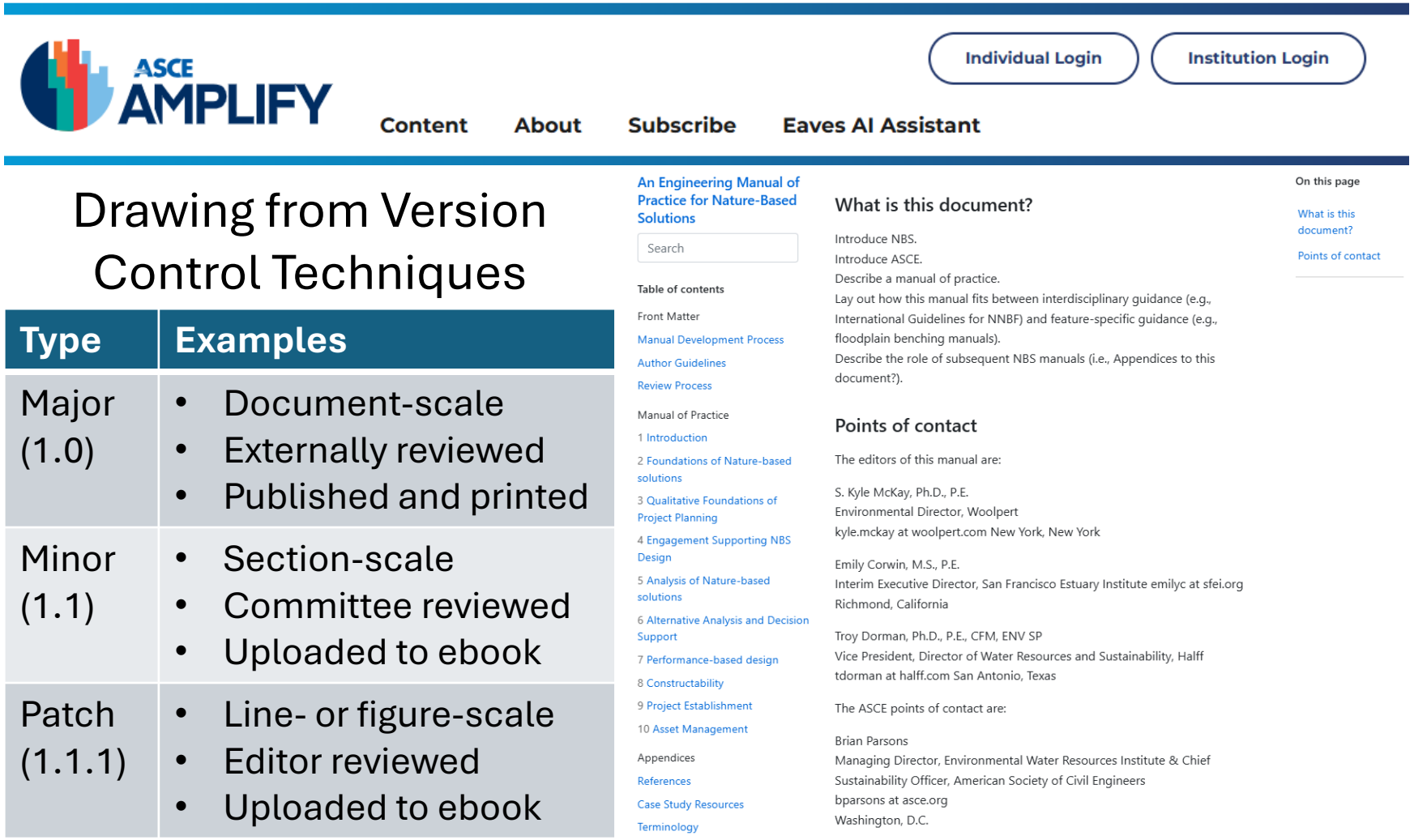
Configuration	Failure Mode	C_W	*a*	*b*	*c*
3-bag	Sliding	0.20	0.48	0.35	0.80
3-bag	Crest Displacement	0.25	0.50	0.20	1.40
5-bag	Sliding	0.30	0.40	0.40	1.40
5-bag	Crest Displacement	0.60	0.50	0.20	3.70

How will the manual be developed?

- ASCE's first fully open access manual
- Design the manual for adaptability at multiple time scales
- Modular content with an emphasis on internal cross-referencing
- Led by engineers with input from other experts and disciplines

Drawing from Version Control Techniques

Type	Examples
Major (1.0)	• Document-scale • Externally reviewed • Published and printed
Minor (1.1)	• Section-scale • Committee reviewed • Uploaded to ebook
Patch (1.1.1)	• Line- or figure-scale • Editor reviewed • Uploaded to ebook



Scoping the manual of practice

- Audience of new and seasoned professionals
  - Primary = design engineers
  - Second = designers in other fields
- Geography
  - Not explicitly tied to a location
  - Acknowledging international use of ASCE products
- Sectors of application
  - Principles and methods agnostic to sector
  - Define a problem-centric view of how NBS are used
- Include common NBS topics, but emphasize an engineer's roles and responsibilities
- What is NOT included?
  - Technical guidelines for specific features
  - Broad changes to general guidance on engineering practice (e.g., cost estimation, cost-benefit analysis)





Photos from Webb et al. (2019, FHWA-HEP-19-042)

What is in the manual?

CHAPTERS	REPRESENTATIVE THEMES
Introduction and purpose	Need for engineering-oriented NBS guidance, audience, geographic scope
Foundations of NBS	Defining NBS, typology of NBS, existing guidance, design principles
Qualitative foundations of project design	Systems context, primary and secondary design objectives, risk tolerance, constraints, design maturity targets, site assessment
Engagement and positionality	Project management, interdisciplinary dialog, ways of knowing, ladder of participation, facilitation, co-production
Scoping NBS Analysis	Levels of effort, assessing outcomes/benefits, project performance, monetary cost, non-monetary cost
Alternative analysis	Management measures, alternative development, decision support, trade-off analysis
Performance-based design	Iterative workflows, integrated modeling, hazard resilience, reliability, adaptability
Constructability	Material selection, regulatory requirements, developing specifications, construction oversight
Adaptive management	Compliance monitoring, performance monitoring, adaptive management, transition to operations
Asset Management	Condition assessment and inspection, operations, maintenance, repair, rehabilitation, decommissioning, life cycle budgeting

What is the timeline for development?

ASCE Inspire (2023)

Articulating a need

Conference workshop Alexandria, VA

NBS Task Force founded (2024)

Guiding the ASCE process

Developing proposals for ASCE

ISLE Learning Series (2025)

Monthly webinars

Scoping the manual

Building a community

(600 attendees!)

Gathering the forces (2026)

Identifying the manual team (editors, review panel, chapter leads, authors)

Finding the resources

Drafting manual contents (2027)

Synopses by chapter leads

Alignment of chapters at ASCE2027

Contributors draft chapters

Making it a manual (2028)

Editing chapters

External peer-review

Response to comment

Publication on ASCE site

Who is involved in manual development?

ROLE	DESCRIPTION
EWRI Technical Committee	• Long-term standing advancing NBS in ASCE (policy statement) • Lead society-wide initiatives on education, policy, standards, etc.
NBS Task Force	• Initiates manual development (scoping, securing funding, etc.) • Initiates and coordinates other NBS products (e.g., other manuals)
ASCE Staff (Parsons, Hocker, Snyder)	• Coordinates and connects across the society • Serves as lead granting entity on proposals • Provides support on file sharing, publishing models, etc.
ASCE Executive Committee	• Oversees MOP development writ large • Approves editors and blue ribbon panel • Ensures quality of the review, response, and revision processes
Editors (Corwin, Dorman, McKay)	• Develops scope and outline • Leads coordination across executive committee, author teams, and blue ribbon panel • Identifies chapter leads and oversees content development • Aligns content and voice across chapters
Project Managers (Libby, Del Monaco)	• Facilitates author team meetings • Develops author guidance • Coordinates logistics with ASCE staff and editorial team
Blue Ribbon Panelists (pending approval)	• Reviews outline, first-order drafts, and response to comments
Chapter Leads (1 Practice and 1 Research Lead)	• Leads compilation of chapter • Coordinates content with Editors • Responds to comments and revisions
Chapter Contributors	• Contributes content as directed by the chapter leads • Gathers content and methods from audiences not engaged in MOP development

Paper Discussion

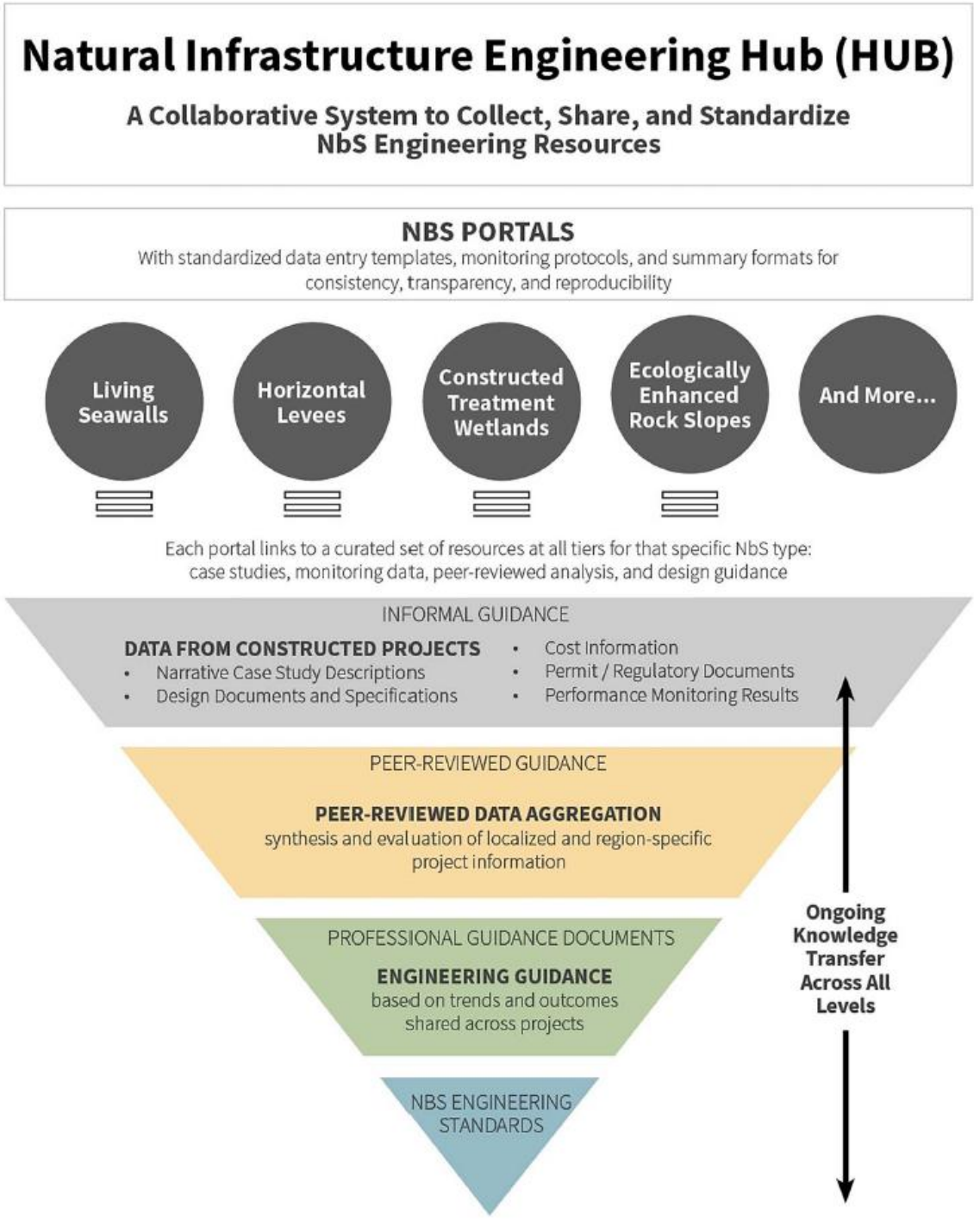
Bredes, A., Tso, G., Gittman, R.K., Narayan, S., Tomiczek, T., Miller, J.K. and Morris, R.L., 2024. A 20-year systematic review of wave dissipation by soft and hybrid nature-based solutions (NbS). Ecological Engineering, 209, p.107418.

Corwin, E., Reguero, B.G., McKay, S.K., Piercy, C.D., Pontee, N.I., Tomiczek, T., Jameson, T., Narayan, S., Sridharan, V.K. and Bledsoe, B.P., 2026. Nature-Based Design Standards: Past, Present, and Future. ASCE OPEN: Multidisciplinary Journal of Civil Engineering, 4(1), p.04026002.

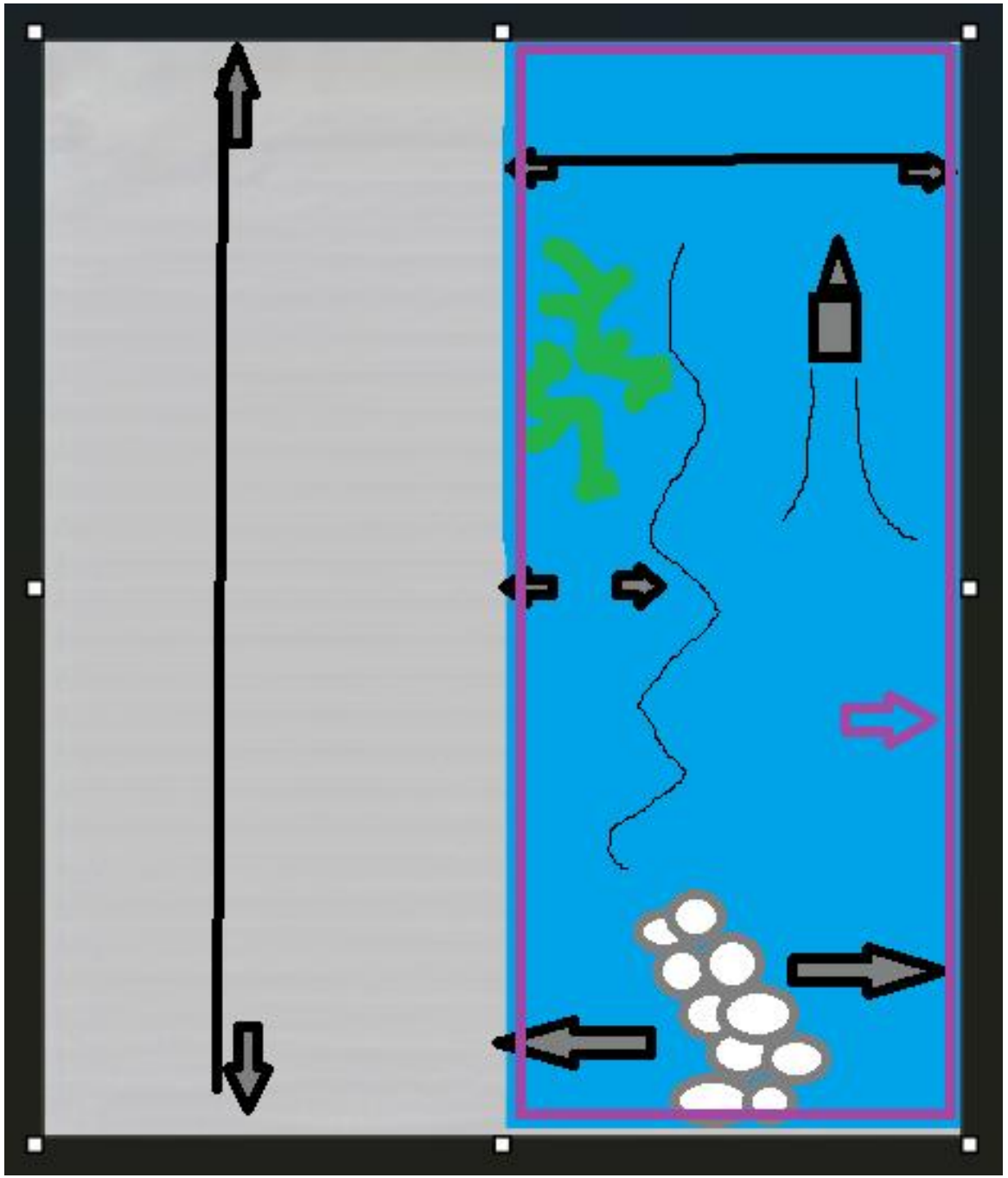
Esquivel-Muelbert, J.R., Fontoura, L., Zawada, K., Erickson, K., Figueira, W., Madin, J.S. and Bishop, M.J., 2026. The natural architecture of oyster reefs maximizes recruit survival. Nature, pp.1-5.

Corwin et al. (2026)

- Will we ever get to “standards” for NBS design?
- Is a “hub” needed? How to attract users and build content?
  - What should bracket the scope of an NI Hub?
  - How should NI features be organized?
  - What features are ready for this type of Hub?
  - What content should the Hub be built around?

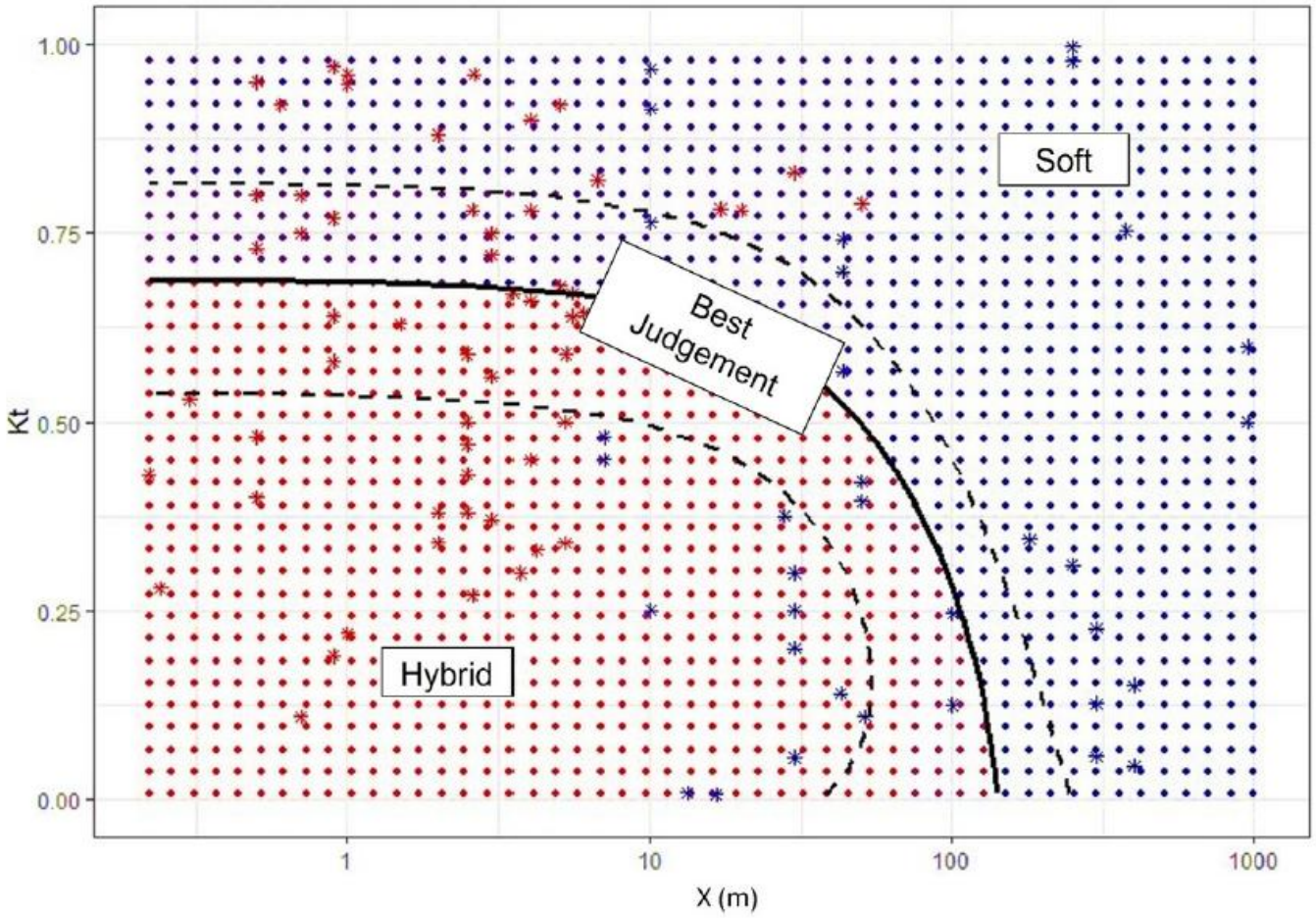


Bredes et al. (2024)



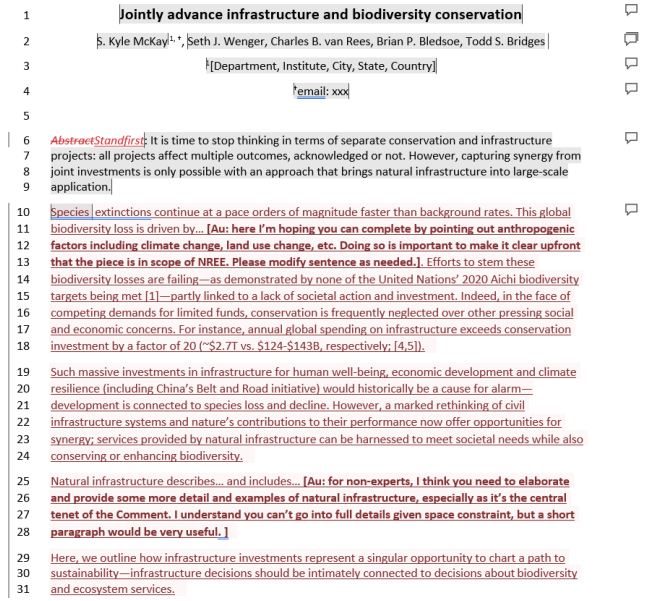
Bredes et al. (2024)

- Synthesis vs. systematic review vs. meta-analysis
- Revisiting inclusion criteria
- Space limitations for different techniques
- Prioritization of monitoring metrics – relative to what?
- What is the US Coastal Research Program?



Esquivel-Muelbert et al. (2026)

- Ruthless editing at high-impact journals
- Robust experimental design
- Marrying basic and applied findings in one paper
- How would these findings influence design?
- Do the authors offer a playbook for design processes?



Jointly advance infrastructure and biodiversity conservation

S. Kyle McKay[1, *], Seth J. Wenger, Charles B. van Rees, Brian P. Bledsoe, Todd S. Bridges

[1][Department, Institute, City, State, Country]

[*]email: xxx

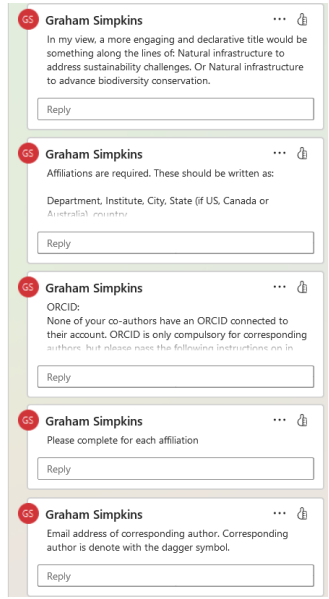
~~Abstract~~*Standfirst*: It is time to stop thinking in terms of separate conservation and infrastructure projects: all projects affect multiple outcomes, acknowledged or not. However, capturing synergy from joint investments is only possible with an approach that brings natural infrastructure into large-scale application.

Species extinctions continue at a pace orders of magnitude faster than background rates. This global biodiversity loss is driven by… **[Au: here I'm hoping you can complete by pointing out anthropogenic factors including climate change, land use change, etc. Doing so is important to make it clear upfront that the piece is in scope of NREE. Please modify sentence as needed.]**. Efforts to stem these biodiversity losses are failing—as demonstrated by none of the United Nations' 2020 Aichi biodiversity targets being met [1]—partly linked to a lack of societal action and investment. Indeed, in the face of competing demands for limited funds, conservation is frequently neglected over other pressing social and economic concerns. For instance, annual global spending on infrastructure exceeds conservation investment by a factor of 20 (~$2.7T vs. $124-$143B, respectively; [4,5]).

Such massive investments in infrastructure for human well-being, economic development and climate resilience (including China's Belt and Road initiative) would historically be a cause for alarm—development is connected to species loss and decline. However, a marked rethinking of civil infrastructure systems and nature's contributions to their performance now offer opportunities for synergy; services provided by natural infrastructure can be harnessed to meet societal needs while also conserving or enhancing biodiversity.

Natural infrastructure describes… and includes… **[Au: for non-experts, I think you need to elaborate and provide some more detail and examples of natural infrastructure, especially as it's the central tenet of the Comment. I understand you can't go into full details given space constraint, but a short paragraph would be very useful.]**

Here, we outline how infrastructure investments represent a singular opportunity to chart a path to sustainability—infrastructure decisions should be intimately connected to decisions about biodiversity and ecosystem services.



Graham Simpkins
In my view, a more engaging and declarative title would be something along the lines of: Natural infrastructure to address sustainability challenges. Or Natural infrastructure to advance biodiversity conservation.
Reply

Graham Simpkins
Affiliations are required. These should be written as:

Department, Institute, City, State (if US, Canada or Australia), country
Reply

Graham Simpkins
ORCID:
None of your co-authors have an ORCID connected to their account. ORCID is only compulsory for corresponding authors, but please pass the following instructions on in
Reply

Graham Simpkins
Please complete for each affiliation
Reply

Graham Simpkins
Email address of corresponding author. Corresponding author is denote with the dagger symbol.
Reply

Algal processing in exclosures with Freshwater Shrimp (Pringle et al.)



Next Week

Oct 27th papers (McKay is out):

Abera L.E., Surbeck C.Q., and McKay S.K. 2024. Lifecycle cost and benefit analysis for parcel-scale implementation of green stormwater infrastructure. Green Technologies and Sustainability (100139) https://doi.org/10.1016/j.grets.2024.100139.

Chambers M.L., Santiago-Collazo F.L., Lammers R.W., and Bledsoe B.P. 2026. A practitioner-friendly approach to quantifying the hydrologic benefits of city-scale distributed green stormwater infrastructure. Journal of Environmental Management, 409 (130012), https://doi.org/10.1016/j.jenvman.2026.130012.

Webber, M.K., Mei, L. and Samaras, C., 2025. Bridging the gap: riverine nature-based solutions for climate resilient transportation infrastructure in the United States. npj Urban Sustainability, 5(1), p.28.